

Periodic two-letter ringsets in the Lipschitz quaternions: the general odd-period classification

Carptopus
carptopus@163.com

24 August 2026

Public Beta v0.1. The theorem and the complete manuscript have passed independent internal zero-trust audits. External mathematical review and journal peer review are pending.

Abstract

Let $\mathbf{L} = \mathbb{Z}[\mathbf{i}, \mathbf{j}, \mathbf{k}]$ be the ring of Lipschitz quaternions. For a periodic word $w : \mathbb{Z} \rightarrow \{\mathbf{i}, \mathbf{j}\}$ of an arbitrary odd period m , put

$$S_w = \{a + w(a) : a \in \mathbb{Z}\} \subseteq \mathbf{L}.$$

We classify exactly when S_w is a ringset, meaning that the rational quaternion polynomials integer-valued on S_w are closed under multiplication. The criterion is that w is nonconstant and, for every prime $p \mid m$ with $p \equiv 3 \pmod{4}$, every position class modulo $p^{v_p(m)}$ is bichromatic. This removes the squarefreeness hypothesis from the preceding odd-period classification.

The sufficiency direction combines split prime-power cores, paired fibres at nonsplit primes, periodic congruence orbits, and the Chinese remainder theorem. For necessity, a monochromatic class modulo p^ν is detected in $\mathbf{L}/p^{V+1}\mathbf{L}$, where $V = v_p((p^\nu - 1)!)$. A falling-factorial product of central quadratic minimal polynomials gives an explicit null polynomial whose coefficientwise right multiple is non-null. Thus the higher prime-power condition is both necessary and sufficient; it is not inferred from finite computation.

1 Introduction

Let $\mathbb{D}$ be the rational Hamilton quaternion algebra and

$$\mathbf{L} = \mathbb{Z}[\mathbf{i}, \mathbf{j}, \mathbf{k}], \quad \mathbf{i}^2 = \mathbf{j}^2 = -1, \quad \mathbf{ij} = \mathbf{k} = -\mathbf{ji}.$$

The variable in $\mathbb{D}[x]$ is central, and polynomials are evaluated on the right: if $f(x) = \sum_d c_d x^d$, then $f(\alpha) = \sum_d c_d \alpha^d$. For $S \subseteq \mathbf{L}$, define

$$\text{Int}(S, \mathbf{L}) = \{f \in \mathbb{D}[x] : f(S) \subseteq \mathbf{L}\}.$$

Following Werner [5], S is a *ringset* if $\text{Int}(S, \mathbf{L})$ is a subring of $\mathbb{D}[x]$. In a noncommutative coefficient ring this closure is not automatic.

Werner [5] classified the finite ringsets in the Lipschitz and Hurwitz quaternion rings and established residue-null-ideal tools for infinite sets. A preceding paper [1] applied these tools to periodic two-letter sets and obtained a complete criterion for odd squarefree periods. The squarefree hypothesis entered at a precise point: when $p \mid m$, it forced

$$\gcd(p^e, m) = p$$

for every $e \geq 1$. If $p^2 \mid m$, a residue fibre can instead remember a higher position class, and a finite-field obstruction no longer detects all failures.

The purpose of this paper is to close that boundary for every odd period. Let $m \geq 1$ be odd, let $w : \mathbb{Z} \rightarrow \{\mathbf{i}, \mathbf{j}\}$ satisfy $w(a + m) = w(a)$, and put

$$S_w = \{a + w(a) : a \in \mathbb{Z}\}.$$

The displayed period is not required to be least.

Theorem 1.1. *Let m be odd and let $w : \mathbb{Z} \rightarrow \{\mathbf{i}, \mathbf{j}\}$ have period m . Then S_w is a ringset of $\mathbf{L}$ if and only if both of the following conditions hold:*

1. *w is nonconstant;*
2. *for every prime $p \mid m$ with $p \equiv 3 \pmod{4}$, if $\nu = v_p(m)$, then*

$$\{w(a) : a \equiv c \pmod{p^\nu}\} = \{\mathbf{i}, \mathbf{j}\} \quad (c \in \mathbb{Z}/p^\nu\mathbb{Z}).$$

When m is squarefree, $\nu = 1$ and Theorem 1.1 specializes exactly to the criterion in [1]. The new content is the necessity and sufficiency of the highest prime-power position classes when the period is not squarefree.

Corollary 1.2. *If every prime divisor of the odd period m is congruent to 1 modulo 4, then every nonconstant binary word of period m produces a ringset S_w .*

The proof does not claim as new the residue criterion [3, 5], finite matrix-ring core theory [4, 6], fixed-divisor methods [2], or the split and paired-fibre lemmas already used in [1]. The new ingredient is an explicit higher-prime-power obstruction tailored to a monochromatic periodic cylinder.

2 Null ideals and local tools

For a ring A and $T \subseteq A$, let

$$\mathcal{N}(T, A) = \{f \in A[x] : f(t) = 0 \text{ for every } t \in T\}.$$

This is always a left ideal. The set T is *core* if $\mathcal{N}(T, A)$ is a two-sided ideal.

We use the following residue criterion, which follows from Werner [5, Lemmas 5.3 and 5.4].

Lemma 2.1 (Residue criterion). *Let $S \subseteq \mathbf{L}$, and let $\overline{S}_n$ be its image in $\mathbf{L}/n\mathbf{L}$.*

1. If $\overline{S}_n$ is core for every $n \geq 2$, then S is a ringset.
2. If, for some n , there are $g \in \mathcal{N}(\overline{S}_n, \mathbf{L}/n\mathbf{L})$ and $u \in \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ such that $gu \notin \mathcal{N}(\overline{S}_n, \mathbf{L}/n\mathbf{L})$, then S is not a ringset.

Here gu means uniform right multiplication of all coefficients by u .

We also use two local core lemmas from [1]. They are restated to make the argument self-contained.

Lemma 2.2 (Paired fibres). *Let $q \geq 2$. Any subset*

$$T \subseteq \{a + \mathbf{i}, a + \mathbf{j} : a \in \mathbb{Z}/q\mathbb{Z}\}$$

that contains both letters above every real residue is core in $\mathbf{L}/q\mathbf{L}$.

Proof. For a fixed real residue a , divide a null polynomial f by the central minimal polynomial

$$\mu_a(x) = x^2 - 2ax + a^2 + 1.$$

The remainder is $r(x) = \gamma_1 x + \gamma_0$. Its vanishing at $a + \mathbf{i}$ and $a + \mathbf{j}$ gives $\gamma_1(\mathbf{i} - \mathbf{j}) = 0$, hence $2\gamma_1 = 0$ because $(\mathbf{i} - \mathbf{j})(-\mathbf{i} + \mathbf{j}) = 2$. For any quaternion basis unit u and either root α ,

$$(ru)(\alpha) - r(\alpha)u = \gamma_1(u\alpha - \alpha u) = 0,$$

since the commutator is divisible by 2. Centrality of μ_a passes this from the remainder to f . Thus the null ideal is stable under right multiplication by constants, and therefore by all polynomials because it is a left ideal and x is central. $\square$

Lemma 2.3 (Split one-letter fibres). *Let $p \equiv 1 \pmod{4}$, $e \geq 1$, and $R = \mathbb{Z}/p^e\mathbb{Z}$. If*

$$T \subseteq \{a + \mathbf{i}, a + \mathbf{j} : a \in R\}$$

contains at least one point above every real residue, then T is core in $\mathbf{L}/p^e\mathbf{L}$.

Proof. Choose $s \in R^\times$ with $s^2 = -1$. The standard representation

$$\mathbf{i} \mapsto \begin{pmatrix} s & 0 \\ 0 & -s \end{pmatrix}, \quad \mathbf{j} \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

identifies $\mathbf{L}/p^e\mathbf{L}$ with $M_2(R)$. For $u \in \{\mathbf{i}, \mathbf{j}\}$, put

$$e_u^+ = \frac{1 + u/s}{2}, \quad e_u^- = \frac{1 - u/s}{2}.$$

These are complementary orthogonal idempotents and

$$(a + u)^d = (a + s)^d e_u^+ + (a - s)^d e_u^-.$$

The image of a plus idempotent for either letter and the image of a minus idempotent for either letter are complementary rank-one summands. Explicit generators give determinants $1, -s, 1, -2s$ in the four possible pairings, all units in R .

For a null polynomial $f(x) = \sum_d c_d x^d$, let $F(z) = \sum_d c_d z^d$. At a fixed $z \in R$, use a selected point of real part $z - s$ and one of real part $z + s$. Their zero evaluations give $F(z)e_{u_1}^+ = F(z)e_{u_2}^- = 0$, so $F(z) = 0$. Consequently the scalar polynomial for every coefficientwise right multiple fr is $F(z)r = 0$, and the idempotent decomposition shows that fr is null on T . The null ideal is two-sided. $\square$

3 Periodic congruence fibres

Lemma 3.1 (Periodic orbit lemma). *Let w have period m , let $q \geq 1$, and fix $a \in \mathbb{Z}/q\mathbb{Z}$. The positions modulo m attained by integers congruent to a modulo q are*

$$a + q\mathbb{Z} \pmod{m} = \{r \in \mathbb{Z}/m\mathbb{Z} : r \equiv a \pmod{\gcd(q, m)}\}.$$

Proof. The subgroup generated by q in $\mathbb{Z}/m\mathbb{Z}$ is the subgroup of multiples of $\gcd(q, m)$. Translate it by a . $\square$

For $q = p^e$ and $\nu = v_p(m)$, a real residue fibre therefore sees one position class modulo

$$p^{\min(e, \nu)}.$$

If every class modulo p^ν is bichromatic, every coarser class is a union of bichromatic classes and is also bichromatic.

The criterion is independent of using a nonminimal period. Indeed, if the least period is $d \mid m$, then a class modulo $p^{v_p(m)}$ projects onto a class modulo $p^{v_p(d)}$; its set of letters is exactly the latter class's set of letters.

4 A prime-power fixed-divisor lemma

The necessity argument requires a precise valuation dichotomy.

Lemma 4.1. *Let p be a prime, $\nu \geq 1$, and put*

$$N = p^\nu - 1, \quad V = v_p(N!), \quad P(d) = \prod_{r=1}^N (d - r).$$

With the convention $v_p(0) = +\infty$,

$$v_p(P(d)) = \begin{cases} V, & p^\nu \mid d, \\ \geq V + 1, & p^\nu \nmid d. \end{cases}$$

Proof. If $p^\nu \mid d$, then $v_p(d - r) = v_p(r)$ for every $1 \leq r < p^\nu$, and summing gives V .

Otherwise choose $s \in \{1, \dots, N\}$ with $s \equiv d \pmod{p^\nu}$. The factor $d - s$ has valuation at least ν . The nonzero residue classes of the remaining $d - r$ modulo p^ν are the nonzero classes other than s , up to permutation. Hence

$$v_p(P(d)) \geq V - v_p(s) + \nu \geq V + 1,$$

because $v_p(s) \leq \nu - 1$. □

We shall work modulo p^E , where

$$E = V + 1.$$

Notice that $E \geq \nu$: the assertion is immediate for $\nu = 1$, while for $\nu \geq 2$ the factors $p, p^2, \dots, p^{\nu-1}$ already contribute at least $\nu - 1$ to V .

5 A higher nonsplit obstruction

Let $p \equiv 3 \pmod{4}$ and $\nu \geq 1$. Suppose that a position class $c \pmod{p^\nu}$ is monochromatic. We construct a finite quotient witness without assuming any classification of subsets over $\mathbb{Z}/p^e\mathbb{Z}$.

Assume first that the class contains only $\mathbf{i}$; the other case is symmetric. Use N, V, E from Section 4 and define, in $(\mathbf{L}/p^E\mathbf{L})[x]$,

$$\mu_{c+r}(x) = (x - (c + r))^2 + 1 \quad (1 \leq r \leq N),$$

$$Q_c(x) = \prod_{r=1}^N \mu_{c+r}(x), \quad h_c(x) = (x - (c + \mathbf{i}))Q_c(x).$$

Every μ_{c+r} , and hence Q_c , has central coefficients.

For $q \in \mathbf{L}$, write

$$v_p(q) = \max\{t \geq 0 : q \in p^t\mathbf{L}\},$$

with $v_p(0) = +\infty$. Equivalently, this is the minimum of the p -adic valuations of the four Lipschitz coordinates. Multiplication by a quaternion whose norm is prime to p preserves this valuation.

Lemma 5.1. *The polynomial h_c is null on the image of S_w modulo p^E , but the coefficientwise right multiple $h_c\mathbf{j}$ is not null.*

Proof. Since $E \geq \nu$ and $\nu = v_p(m)$, we have $\gcd(p^E, m) = p^\nu$. Thus Lemma 3.1 says that every real fibre modulo p^E sees exactly one position class modulo p^ν .

Take a point $y = a + u$ of the periodic image, where $u \in \{\mathbf{i}, \mathbf{j}\}$, and write $d = a - c$. Direct calculation gives

$$\mu_{c+r}(y) = (d - r)(d - r + 2u).$$

The second factor is a unit modulo p^E . Indeed, its norm modulo p is $(d - r)^2 + 4$, which cannot vanish because -1 is not a square when $p \equiv 3 \pmod{4}$. Therefore

$$v_p(Q_c(y)) = v_p(P(d)).$$

If $p^\nu \nmid d$, Lemma 4.1 shows that $Q_c(y) = 0$ modulo p^E . If $p^\nu \mid d$ and $u = \mathbf{i}$, the first factor of $h_c(y)$ is d and the total valuation is at least $V + \nu \geq E$. If $p^\nu \mid d$ and $u = \mathbf{j}$, then the periodic orbit lemma says that the relevant position class is the monochromatic class c , so no such point occurs. Thus h_c is null on the entire periodic image.

Because Q_c is central, coefficientwise right multiplication gives

$$h_c \mathbf{j} = ((x - (c + \mathbf{i}))\mathbf{j})Q_c.$$

At the point $c + \mathbf{i}$ the first factor evaluates, with right evaluation, to

$$\mathbf{j}(c + \mathbf{i}) - (c + \mathbf{i})\mathbf{j} = -2\mathbf{k},$$

a p -adic unit. Lemma 4.1 gives $v_p(Q_c(c + \mathbf{i})) = V$. Hence

$$(h_c \mathbf{j})(c + \mathbf{i}) \not\equiv 0 \pmod{p^{V+1}}.$$

This proves the claim. □

If the monochromatic letter is $\mathbf{j}$, replace the leading factor by $x - (c + \mathbf{j})$ and right-multiply coefficients by $\mathbf{i}$.

6 Proof of Theorem 1.1

Proof. Sufficiency.

Assume conditions 1 and 2. It is enough to show that the image of S_w is core modulo every prime power p^e .

If $p \equiv 1 \pmod{4}$, every real residue occurs and Lemma 2.3 applies; no bichromatic condition is needed.

If $p = 2$, then $\gcd(2^e, m) = 1$ because m is odd. Every real fibre traverses the whole period, so nonconstancy makes it bichromatic. Lemma 2.2 applies.

Now let $p \equiv 3 \pmod{4}$. If $p \nmid m$, the same coprimality argument and Lemma 2.2 apply. If $\nu = v_p(m) > 0$, Lemma 3.1 says that a real fibre modulo p^e sees one position class modulo $p^{\min(e, \nu)}$. Condition 2 makes every class at level p^ν bichromatic, and therefore every coarser class is bichromatic. Lemma 2.2 again applies.

For an arbitrary denominator $n = \prod_r p_r^{e_r}$, the Chinese remainder theorem gives

$$\mathbf{L}/n\mathbf{L} \simeq \prod_r \mathbf{L}/p_r^{e_r}\mathbf{L}.$$

Polynomial coefficients and right evaluation are componentwise, so the null ideal of an image is the product of the null ideals of its projections. Each prime-power projection is core; hence the image modulo n is core. Lemma 2.1 proves that S_w is a ringset.

Necessity.

If w is constant, reduce modulo 3. The finite-field missing-letter obstruction from [1, Lemma 5.1] supplies a null polynomial whose right multiple is non-null. Lemma 2.1 shows that S_w is not a ringset.

Suppose next that condition 2 fails. Choose $p \mid m$ with $p \equiv 3 \pmod{4}$, put $\nu = v_p(m)$, and choose a monochromatic class $c \pmod{p^\nu}$. Lemma 5.1 gives a null polynomial in the image modulo p^{V+1} whose coefficientwise right multiple is non-null. The second part of Lemma 2.1 again shows that S_w is not a ringset. This proves both necessary conditions and completes the theorem. $\square$

7 Consequences and examples

Theorem 1.1 removes the entire odd-period boundary left open in [1]. It also shows exactly why checking classes only modulo p is insufficient when $p^2 \mid m$.

For a displayed period $m = 9$, it is possible for all three classes modulo 3 to be bichromatic. Nevertheless every class modulo 9 consists of a single word position and is therefore monochromatic. Thus no period-9 word satisfies the theorem: choosing any one of these highest classes gives an obstruction modulo 27, since

$$v_3(8!) = 2, \quad E = 3.$$

For period $m = 45 = 3^2 \cdot 5$, only the classes modulo 9 impose a colour condition; the split prime 5 imposes none. For a period divisible only by primes congruent to 1 modulo 4, Corollary 1.2 gives all nonconstant words, regardless of repeated prime factors.

The criterion can be checked using any stated odd period. If a larger nonminimal period is used, the extra prime-power classes merely repeat the classes belonging to the least period, as observed after Lemma 3.1.

8 Computational calibration

The proof is symbolic. Computation was used only to discover and test the prime-power boundary.

- For one coarse class modulo p in the quotient modulo p^2 , each of its p lifts was allowed to contain only **i**, only **j**, or both. For $p = 3$, all $3^3 = 27$ patterns agreed with the local lift-count rule that each letter must occur in at least two lifts. This is weaker than the hypothesis of Lemma 2.2 and is only a finite local-ring observation.
- For $p = 7$ modulo 49, samples agreed with the same lift-count threshold: deleting **j** from one or five lifts remained stable, while deleting it from six or seven lifts did not.
- Explicit obstruction polynomials were evaluated for $(p, \nu) = (3, 1), (3, 2), (7, 1)$.

- An independent row-module computation over $\mathbb{Z}/27\mathbb{Z}$ rejected the finite fibre pattern having only **i** above real residues congruent to 0 modulo 9 and both letters elsewhere. This finite quotient pattern is not itself the residue image of a two-letter word of displayed period 9.

These checks calibrate the formulas but do not prove the general theorem. The arbitrary- p, ν conclusion comes from Lemmas 4.1 and 5.1.

9 Prior-work and novelty boundary

The general ringset/null-ideal link is due to Frisch [3]. Werner [4] classified null ideals of subsets of matrix rings over fields, and Rissner and Werner [6] studied their enumeration. Werner [5] classified finite Lipschitz and Hurwitz quaternion ringsets and established the residue and right-multiplication criteria used here. Peruginelli [2] developed prime-power fixed-divisor ideals in the commutative integer-valued-polynomial setting. None of these general tools is claimed as new.

The preceding paper [1] proves the squarefree instance of Theorem 1.1 and contains the split and finite-field local mechanisms reused here. The present paper claims only the removal of squarefreeness, the exact $p^{v_p(m)}$ condition, and the explicit higher-prime-power obstruction. The prime-square analysis is an intermediate part of this general result, not a separate result of comparable scope.

As of the bounded literature search completed on 23 August 2026, no public source was found that directly states this all-odd-period classification. That assessment is not a certificate of absolute global priority. The result has not received external expert review or formal peer review.

Declaration of generative AI and AI-assisted technologies

During the research and preparation of this work, the author used OpenAI Codex to assist with literature searches, symbolic exploration, exploratory code review, proof checking, and language drafting. The author reviewed and edited all generated output and takes full responsibility for the content of this manuscript.

References

- [1] Carptopus, *Periodic infinite ringsets in the Lipschitz quaternions*, Public Beta v0.1-beta (2026). [doi:10.5281/zenodo.22069614](https://doi.org/10.5281/zenodo.22069614).
- [2] G. Peruginelli, *Primary decomposition of the ideal of polynomials whose fixed divisor is divisible by a prime power*, J. Algebra 398 (2014), 227–242. [doi:10.1016/j.jalgebra.2013.09.016](https://doi.org/10.1016/j.jalgebra.2013.09.016); [arXiv:1304.7450](https://arxiv.org/abs/1304.7450).
- [3] S. Frisch, *Polynomial functions on subsets of non-commutative rings—a link between ringsets and null-ideal sets*, ITM Web Conf. 20 (2018), 01003. [doi:10.1051/itmconf/20182001003](https://doi.org/10.1051/itmconf/20182001003).

- [4] N. J. Werner, *Null ideals of subsets of matrix rings over fields*, Linear Algebra Appl. 642 (2022), 50–72. [doi:10.1016/j.laa.2022.02.007](https://doi.org/10.1016/j.laa.2022.02.007).
- [5] N. J. Werner, *Integer-valued polynomials on subsets of quaternion algebras*, J. Algebra 686 (2026), 195–219. [doi:10.1016/j.jalgebra.2025.08.011](https://doi.org/10.1016/j.jalgebra.2025.08.011); [arXiv:2412.20609v2](https://arxiv.org/abs/2412.20609v2).
- [6] R. Rissner and N. J. Werner, *Counting core sets in matrix rings over finite fields*, Linear Algebra Appl. 720 (2025), 1–25. [doi:10.1016/j.laa.2025.04.006](https://doi.org/10.1016/j.laa.2025.04.006); [arXiv:2405.04106v2](https://arxiv.org/abs/2405.04106v2).